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Symposium on Clinical and Statistical Prediction[1]

The Tie That Binds

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My task this morning is mainly to set the stage for Drs. Tiedeman and McArthur, which requires only a few minutes. Suppose we try to anticipate a fact about an individual's behavior or history, in advance of having direct knowledge of this fact. This is an attempt to *predict,* even if, as in some cases, the behavior itself took place in the past; since we, the investigators, do not know the fact in question, but infer it. There are many kinds of behavioral "facts" which we may desire to predict. They may be narrowly circumscribed "behavioral atoms," such as what word the subject will utter in response to a given stimulus-word; or, they may be long-term, complex behavior-*consequences,* such as whether the subject will be graduated from college.

There are also many different methods by which this prediction attempt can be made; but they can be pretty well sorted into two large classes. The first class of prediction methods I call "statistical." Given any kind of data, a table or formula or nomograph is employed to grind out a prediction. The other class of methods, called "clinical," utilizes the cerebrations of a skilled human judge. There are points to be made for and against each of these two methods, and our speakers will develop some of them this morning.

But whatever else we may say or think, there is one overarching procedure that is always applicable to both. We can keep a record of the hit-frequency—the predictive success—of each method, as applied within a specified population of decision occasions. And we can then ask, "Which method does the best predictive job?" It is foolish, and I would say even immoral, for a trusted, well-paid expert who accepts the responsibility of making humanly important predictions to employ a method which has a lower hit-frequency than another available method. *All* methods must be subjected to the test of success-frequency. This is the "tie that binds." It binds together Freud and Thurstone, Cattell and Allport, Rorschach and Hathaway, Tiedeman and McArthur. It can serve as a basis of communication among psychologists of very different persuasions, because it transcends all personality theories and cuts across all methodological preferences. For the business of theories and methods is to fit the data, which are behaviors.

We do not assume that all statistical methods are equally appropriate for the problems to which they are applied, nor do we assume that all clinicians are equally clever. We do not assume that the relative power of the two methods will be the same in all predictive domains. The empirical evidence to date consists of some two dozen investigations, in none of which the clinician predicts better than the statistician, and in most of which, he predicts worse. (McNemar has pointed out that my book was in error about the single "win" I assigned to the clinician.) As a clinician, I happen to believe that this depressingly uniform trend is largely due to the character of the prediction problems which have thus far been systematically studied with comparative intent. If so, it is very important to clarify the logical and mathematical structure of the two methods, so

[1]Symposium presented at annual meeting of the American College Personnel Association, Washington, D. C., March 28, 1956. The papers are only slightly modified from their original form and, at the editor's request, the Tiedeman and McArthur papers were not changed after the authors had read Meehl's comments. This brings the full flavor of the argument to the reader.

that we can, in the light of an improved theoretical comprehension, design empirical studies more suited to the clinician's allegedly unique powers. Tiedeman and McArthur will attempt to clarify this logical situation further. I may say here that I remain convinced of some kind of fundamental difference between the two methods; but I must admit that in my little book on the subject I did not succeed in formulating precisely what that difference is. Nor, in my opinion, has anyone else.

My own personal hunch is, that clinical inference is a mixture of the kind of inference that goes on in physics, with the kind that goes on in history or archeology. I do not myself see how the recent exciting research on the Dead Sea Scrolls could have been done statistically, as regards the historical reconstructions, hypotheses, and derivations involved (although actuarial methods have of course been utilized as *part* of the armamentarium of Biblical scholars for almost 100 years). Paleographers and archeologists do make predictions. They proceed by rational methods and they utilize certain nomothetic principles in their hypothesis-building. But they seem not to be actuarial in any sense close to the Trait Model which Tiedeman will present. I am inclined to think that if we could only state rigorously both how history *differs* from physics, and how the two are *similar*, we could easily solve the present analytic problem. Unfortunately no one has yet done a good job on the logic of history either.

This morning's discussion should move us further along the way of understanding. Having read the papers, and having had illuminating correspondence with the two chief speakers, I am confident that it will.

The Trait Model

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Prediction

To predict is to declare which of several possible events *will* occur *before* the event or events *do* occur. A prediction is successful when the prediction later coincides with the actual event or events; otherwise a prediction is unsuccessful. Systems of prediction vary in their rates of success. We prefer the prediction system yielding the highest rate of success.

Prediction necessarily occurs in the absence of the event to be predicted. Therefore, in order to predict we designate two time-points within the life experiences of a particular population of people such that the second time-point, t_2 say, always follows the first time-point, t_1, by a specified time interval. t_1 is the time in the lives of members of the population at which we are to predict; t_2 is the time in the lives of these same people at which we are to specify the behavior of the person. We will call the data collected for each member of a population prior to the time t_1 "antecedent information," or, more simply, "antecedents." We will call the events which will be known at the time t_2 "consequent events," or, more simply, "consequents."

From previous discusion with Dr. McArthur, I have found that an important distinction between the trait model and the dynamic model is whether an experience table is called into play in structuring either model or not. In the trait model an experience table must be considered and must be considered early because it serves as the screen for choice among alternative systems of prediction. Consequently, I will discuss this problem next and it should be remembered that by doing so I move from a general discussion of the logic of prediction to a specific discussion of the logic of prediction by means of the trait model.

[1]The author is indebted to Professor Israel Scheffler and to Professor John B. Carroll for helpful criticism of an early draft of this paper. Responsibility for the paper remains that of the author, of course.